

Portfolio Case Study — UNDP M&E Indicator Tracker

Project Overview

The UNDP Kano M&E Indicator Tracker is a professional analytics project designed to simulate a real-world Monitoring & Evaluation dashboard used by development organizations such as UNDP, UNICEF, WFP, and international NGOs.

The goal of this project was to build a centralized reporting solution for tracking programme performance across Kano State, Nigeria.

The Challenge

Large development programmes often generate massive amounts of operational data from field teams, implementing partners, and beneficiaries.

Without centralized reporting:

- * KPI tracking becomes difficult
- * Budget visibility decreases
- * Risk escalation slows down
- * Decision-making becomes reactive instead of proactive

The challenge was to convert fragmented programme data into a single decision-support dashboard.

The Solution

I designed and built a complete Power BI M&E dashboard capable of monitoring:

- * Beneficiary reach
- * Gender inclusion
- * Youth participation
- * Programme performance
- * Budget utilization
- * Operational risks

The solution provides real-time insights to programme managers and decision-makers.

Dataset

A synthetic but realistic dataset was created using Python to simulate actual programme records.

Dataset scope:

- * 1,500+ beneficiary records
- * 12 LGAs in Kano State
- * 4 integrated tables
- * Multiple KPI indicators
- * Budget allocation records
- * Risk register entries

Tools Used

- * Python
- * Excel
- * Power BI
- * Power Query
- * DAX

Technical Process

Data Generation

Used Python to generate realistic UNDP-style programme datasets.

Data Cleaning

Cleaned and standardized records using Excel and Power Query.

Data Modeling

Built relationships between:

- * Beneficiaries
- * KPI Tracker
- * Budget Tracker
- * Risk Register

DAX Calculations

Created measures for:

- * Total beneficiaries
- * Female %
- * Youth %
- * Total grants
- * Budget utilization
- * Risk counts

Dashboard Design

Built 4 interactive dashboard pages.

Dashboard Pages

Executive Overview

High-level KPI summary.

Programme Performance

Target vs actual performance analysis.

Budget Monitoring

Budget allocation and expenditure tracking.

Risk Dashboard

Risk severity and issue tracking.

Key Insights

- * 1,500+ beneficiaries tracked
- * Female participation exceeded 50%
- * Urban LGAs outperformed rural LGAs
- * Budget utilization exceeded 80%
- * Strong post-intervention income growth
- * Most risks were medium severity

Business Impact

The dashboard improves programme governance by enabling:

- * Faster reporting
- * Better accountability
- * Improved fund monitoring
- * Early risk detection
- * Better resource allocation

Skills Demonstrated

This project demonstrates strong capabilities in:

- * Data Analytics
- * Dashboard Development

- * Monitoring & Evaluation

- * KPI Reporting

- * Business Intelligence

- * Data Storytelling

Final Reflection

This project strengthened my ability to design data solutions for international development programmes.

It reflects my capability to transform raw data into actionable insights that support operational excellence and strategic decision-making.